Lap 2

1. Docker/Docker Compose files

```yaml
services:
  redis-node1:
    image: redis:latest
    container_name: redis-node1
    ports:
      - "6379:6379"

  redis-node2:
    image: redis:latest
    container_name: redis-node2
    command: redis-server --replicaof redis-node1 6379
    ports:
      - "6380:6379"
    depends_on:
      - redis-node1

  etcd:
    image: gcr.io/etcd-development/etcd:v3.5.0
    container_name: etcd
    command:
      - etcd
      - --advertise-client-urls=http://0.0.0.0:2379
      - --listen-client-urls=http://0.0.0.0:2379
    ports:
      - "2379:2379"
```

```
PS E:\cmc-lab-02> notepad E:\cmc-lab-02\docker-compose.yml
PS E:\cmc-lab-02>
PS E:\cmc-lab-02> docker-compose up -d
[+] up 22/22
 ✔Image redis:latest                        Pulled                                    27.0s
 ✔Image gcr.io/etcd-development/etcd:v3.5.0 Pulled                                    36.7s
 ✔Network cmc-lab-02_default                Created                                    0.1s
 ✔Container redis-node1                     Started                                    1.8s
 ✔Container etcd                            Started                                    1.8s
 ✔Container redis-node2                     Started                                    1.7s
```

2. Redis and etcd commands/results

SET mykey "hello" on redis-node1

GET mykey on redis-node2 (showing replication works)

```
PS E:\cmc-lab-02> docker exec -it redis-node1 redis-cli SET mykey "hello"
OK

What's next:
    Try Docker Debug for seamless, persistent debugging tools in any container or image → docker debug redis-node1
    Learn more at https://docs.docker.com/go/debug-cli/
PS E:\cmc-lab-02> docker exec -it redis-node2 redis-cli GET mykey
"hello"
```

docker stop redis-node2 (partition simulation)

```
PS E:\cmc-lab-02> docker stop redis-node2
redis-node2
PS E:\cmc-lab-02> docker exec -it redis-node1 redis-cli SET partitionkey "written-during-partition"
OK
```

docker start redis-node2

```
PS E:\cmc-lab-02> docker start redis-node2
redis-node2
PS E:\cmc-lab-02> docker exec -it redis-node2 redis-cli GET partitionkey
"written-during-partition"

What's next:
    Try Docker Debug for seamless, persistent debugging tools in any container or image → docker debug redis
-node2
    Learn more at https://docs.docker.com/go/debug-cli/
```

etcdctl put foo bar and etcdctl get foo

```
PS E:\cmc-lab-02> docker exec -it etcd etcdctl put foo bar
OK

What's next:
    Try Docker Debug for seamless, persistent debugging tools in any container or image → docker debug etcd
    Learn more at https://docs.docker.com/go/debug-cli/
PS E:\cmc-lab-02> docker exec -it etcd etcdctl get foo
foo
bar
```

etcdctl endpoint status (before and after stopping etcd)

```
PS E:\cmc-lab-02> docker exec -it etcd etcdctl endpoint status
127.0.0.1:2379, 8e9e05c52164694d, 3.5.0, 20 kB, true, false, 2, 5, 5,
```

3. Replication, partition, leader election, failover

```
PS E:\cmc-lab-02> docker-compose down
[+] down 4/4
 ✔Container etcd              Removed
        0.6s
 ✔Container redis-node2       Removed
        0.5s
 ✔Container redis-node1       Removed
        0.5s
 ✔Network cmc-lab-02_default Removed
        0.3s
```